

Static SLSQP / Lookup Table

Current inflow snapshot (ϕ_t, v_t)



Offline optimisation $\max_y P_{\text{farm}}$



Static yaw vector γ_t^*

No wake-propagation delay

No actuator rate constraint

No yaw-activity cost

No fail-safe fallback

temporal context

+ supervisory wrapper

Deployment Gap

Wake-propagation delay (~90-110 s)

Evolving inflow (AR(1) variability)

Actuator rate limits (0.3-0.5 deg/s)

Yaw-activity cost (bearing/gearbox wear)

DRL-Deploy (this work)

SCADA wind + J-step observation history (150 s)



PPO policy MLP (0.1 ms)

Supervisory Wrapper



Gate | Hysteresis | Deadband | Rate limit | Fallback



Actuator-compatible yaw command γ_t